

Stretch, fold, mod out

expanders on Cantor cylinders, and what a ReLU network does to a profinite structure

Working note III for the continuation-algebra programme
self-reviewed draft, not independently verified

3 October 2026

Abstract

We make the abstraction explicit. The object is not the weights or hidden vectors as such. It is (i) a Cantor set with a *profinite structure* (Pearson–Bellissard, PB, Def. 7: a nested family of “mod out at scale ε ” equivalence relations, equivalently a Michon tree), (ii) a finite family of transformations compatible with it, and (iii) a state.

The expander construction recalled from the lecture (vertices are pairs of residues, a few affine maps, local neighbourhood analysis, then a global spectral gap) fits this exactly. The Margulis–Gabber–Galil graphs on $(\mathbb{Z}/p^n)^2$ are the level- n cylinder quotients of one affine action on the p -adic plane $\mathbb{Z}_p^2$, a regular ultrametric Cantor set in PB’s sense. The uniform second-eigenvalue bound $5\sqrt{2}/8$ is expansion at every level of the Michon tree at once. From it we obtain an explicit, constant-degree (eight maps per level) operator family sparsifying PB-type hierarchical diffusion. It commutes with every refinement level, and its Dirichlet form is equivalent to the PB form with constants $(1 - 5\sqrt{2}/8, 2)$.

On the network side, a ReLU layer is *stretch* (linear) followed by *fold* (the sign quotient $z \sim -z$). The network induces pseudometrics on a source Cantor set. PB’s ε -chains turn these into profinite structures, and Lipschitz layers act on them as tree maps that *only merge*. This is the precise sense in which “the MLP gradually mods out”. For random He-initialized networks the stretch is critical: fine scales are kept, to first order, while coarse scales are compressed toward $s_\ell \approx 3\pi/\ell$. On a Cantor-structured input the coarsest separation falls from 1.30 to 0.35 over 16 layers while the finest stays at 0.017, and ε -chain classes merge from 8 to 2. A Galton–Watson trichotomy (sub-, exactly, super-critical) classifies when such dynamics expand. Random ReLU nets sit exactly at criticality, which is the abstract reason the network itself never supplies a spectral gap. Expanders must enter through the transverse/input side.

1 The object, stated abstractly

Definition 1.1 (PB, §4.1 and Def. 7). Let (C, d) be a Cantor set with a regular (pseudo)metric. An ε -chain from x to y is $x = x_0, \dots, x_m = y$ with $d(x_i, x_{i+1}) < \varepsilon$. The relation $x \stackrel{\varepsilon}{\sim} y$ (“equal modulo scale ε ”) is an open equivalence relation with finitely many classes, the *cylinders at scale ε* . The *separation* $\delta(x, y) = \inf\{\varepsilon : x \stackrel{\varepsilon}{\sim} y\}$ is the maximal ultrametric dominated by d (PB Prop. 4). The family $\{\stackrel{\varepsilon}{\sim}\}_{\varepsilon>0}$ is a *profinite structure*. By Michon’s correspondence (PB Props. 5–6) it is the same thing as a reduced weighted rooted tree with weight = cylinder diameter.

Definition 1.2. A map $\varphi : C \rightarrow C$ is λ -compatible if $x \stackrel{\varepsilon}{\sim} y \Rightarrow \varphi x \stackrel{\lambda\varepsilon}{\sim} \varphi y$, equivalently $\delta(\varphi x, \varphi y) \leq \lambda \delta(x, y)$. It then descends to a map of finite sets $C/\stackrel{\varepsilon}{\sim} \rightarrow C/\stackrel{\lambda\varepsilon}{\sim}$.

Proposition 1.3 (Lipschitz maps descend). *If $d'(\varphi x, \varphi y) \leq \lambda d(x, y)$, then $\delta'(\varphi x, \varphi y) \leq \lambda \delta(x, y)$: an ε -chain maps to a $\lambda\varepsilon$ -chain.*

Three kinds of compatible transformations will matter:

- **Contractions** ($\lambda < 1$) generate the tree. The triadic Cantor set is the attractor of the *two* maps $x \mapsto x/3$ and $x \mapsto x/3 + 2/3$. These are the Cuntz isometries s_0, s_1 of note I, §7, and the “two transformations” that produce children of a node.
- **Isometries** ($\lambda = 1$, bijective) act inside each level and give a *Schreier graph* on the cylinders at every scale. This is where expanders live.
- **Expansions** ($\lambda > 1$), such as the shift, forget the top digit. Their transfer operators carry mixing.

The *state* is a measure on C , e.g. the gain-weighted law of the continuation note. “Modding out” is passing to a cylinder quotient. “Expansion” is a spectral gap of the transformations’ action that is uniform over all scales.

2 The lecture example, made exact inside PB

Let $C = \mathbb{Z}_p^2$ (the p -adic plane) with the ultrametric $\max(|x|_p, |y|_p)$. Its cylinders at level n are the residue classes modulo p^n , so “mod p^n ” is PB’s modding out at scale p^{-n} . Its Michon tree has p^2 children per vertex, and Haar measure is the uniform state. The eight Gabber–Galil maps

$$(x, y) \mapsto (x, y \pm 2x), (x, y \pm (2x + 1)), (x \pm 2y, y), (x \pm (2y + 1), y)$$

are affine with unimodular linear part. They are therefore Haar-preserving *isometries* that permute the cylinders of every level. The level- n Schreier graph is the Gabber–Galil graph on $(\mathbb{Z}/p^n)^2$.

Theorem 2.1 (Gabber–Galil; uniform over levels). *For every m , the second eigenvalue of the normalized 8-regular Gabber–Galil graph on $(\mathbb{Z}/m)^2$ is at most $\lambda^* = 5\sqrt{2}/8 \approx 0.884$. Hence the Markov operator $M = \frac{1}{8} \sum_{\varphi} U_{\varphi}$ on $L^2(\mathbb{Z}_p^2)$ commutes with every level expectation E_n and has spectrum in $[-1, \lambda^*]$ on L_0^2 . In other words, the profinite action has a spectral gap, the profinite form of “expansion at every level”. (This correspondence between expanding finite quotients and spectral gap of the boundary action is standard; cf. Lubotzky, Abért–Elek.)*

Remark 2.2 (local to global, in PB’s language). Gabber–Galil’s proof works on the dual lattice. Characters of $\mathbb{Z}_p^2$ are indexed by $(\mathbb{Q}_p/\mathbb{Z}_p)^2$, and a character’s *depth* is the p -power of its denominator: exactly the depth bands of note I. The dual maps preserve depth. The local lemma says that for every nonzero frequency, a weighted sum over its images under the generators exceeds its own weight by a fixed factor: a neighbourhood statement. Summed over a band, this gives the global gap. So the expander mechanism is *intra-band* mixing. PB’s heat, by contrast, is *inter-band* damping. The two are complementary.

Theorem 2.3 (explicit constant-degree hierarchical sparsifier). *For $v = r + p^n w$ (r a residue mod p^n , $w \in \mathbb{Z}_p^2$), define the tail operator $M_n f(v) = \frac{1}{8} \sum_{\varphi} f(r + p^n \varphi(w))$. Then:*

1. M_n is a self-adjoint Markov operator. It fixes level- n functions, preserves $L^2(\text{level } m)$ for every m , and commutes with every E_m .
2. $(1 - \lambda^*)(I - E_n) \leq I - M_n \leq 2(I - E_n)$.
3. For any $0 = \lambda_0 < \lambda_1 < \dots$, the generator $\hat{L} = \sum_n (\lambda_{n+1} - \lambda_n)(M_n - I)$ generates a symmetric Markov semigroup with $(1 - \lambda^*)Q_{PB} \leq \hat{Q} \leq 2Q_{PB}$, where $Q_{PB}(f) = \sum_n (\lambda_{n+1} - \lambda_n) \langle f, (I - E_n)f \rangle$. Its gap is $\geq (1 - \lambda^*)\lambda_1$, and it is refinement-compatible.

4. It uses 8 explicit maps per level: $8N$ for N levels, against the $O(N^2/\varepsilon^2)$ random jumps of note 1's small-bias construction. The price is constant-factor rather than $(1 \pm \varepsilon)$ spectral equivalence.

Proof. (1) On each level- n cylinder, M_n is the Gabber–Galil operator in the rescaled coordinate w . Affine maps with integer coefficients preserve residues of w modulo every p^k , hence the level- $(n+k)$ partitions, and Haar measure. The generator set is closed under inverses. A self-adjoint operator preserving a closed subspace commutes with its orthogonal projection. (2) On the orthocomplement of level- n functions, M_n acts on each cylinder as the Gabber–Galil operator on mean-zero functions of w . These are dense in the union over k of functions of $w \bmod p^k$, where the spectrum is in $[-1, \lambda^*]$. (3) Sum (2) with nonnegative weights; each inequality holds termwise as quadratic forms. $\square$

Numerics. The second eigenvalues of the walk on $(\mathbb{Z}/p^n)^2$ all stay below $\lambda^* = 0.884$:

p	modulus $m = p^n$: second eigenvalue
2	2: 0.500, 4: 0.604, 8: 0.703, 16: 0.757, 32: 0.787, 64: 0.807, 128: 0.820
3	3: 0.576, 9: 0.731, 27: 0.783, 81: 0.812
5	5: 0.666, 25: 0.781, 125: 0.820
7	7: 0.722, 49: 0.801

For $p = 2$, $N = 6$, $n = 2$, the tail operator fixes level-2 functions exactly, $[M_n, E_n] = 0$, and its spectrum on the complement is $[-0.734, 0.757]$.

3 The network as stretch-and-fold on a profinite structure

3.1 Fold = mod out by a sign

$\text{relu}(z) = \frac{1}{2}(z + |z|)$, and $|\cdot|$ is the quotient $\mathbb{R} \rightarrow \mathbb{R}/\{\pm 1\}$. A layer is a linear *stretch* followed by a coordinatewise *fold*, and its activation bit records which branch was taken.

Example 3.1 (the tent map is a ReLU layer and acts as the shift). $T(x) = 2\min(x, 1-x) = 2\text{relu}(x) - 4\text{relu}(x - \frac{1}{2})$ on $[0, 1]$ is a width-2 ReLU layer. Its itinerary (which half at each step) is a binary address (Gray-coded binary expansion). T acts on addresses as the shift: a 2-compatible expansion. Depth- L compositions are Telgarsky’s sawtooths with 2^L pieces, i.e. the level- L cylinders. Stretch 2 gives exponential mixing (a spectral gap of the transfer operator on Lipschitz/BV observables). So a stretch-and-fold network *can* be an expanding, gapped system.

3.2 Network-induced profinite structures

Fix a source Cantor set C with a continuous coding $\iota : C \rightarrow S^{n-1}$ of input directions (e.g. the fair binary source of the continuation note). Each depth ℓ gives a pseudometric $d_\ell(\xi, \eta) = \|h_\ell(\iota\xi) - h_\ell(\iota\eta)\|$, or its angular version. PB’s ε -chains turn it into a profinite structure $\{\tilde{\varepsilon}_\ell\}$ with separation δ_ℓ and Michon tree $\mathcal{T}_\ell$. Points with $\delta_\ell = 0$ are identified outright; this happens, for instance, when all of a layer’s bits switch off.

Proposition 3.2 (the MLP only merges). $d_{\ell+1} \leq \|W_{\ell+1}\|_{op} d_\ell$ because ReLU is 1-Lipschitz. Hence $\delta_{\ell+1} \leq \|W_{\ell+1}\| \delta_\ell$, and each class of $\tilde{\varepsilon}_\ell$ lies inside a class of $\tilde{\varepsilon}_{\ell+1}$. The trees $\mathcal{T}_0 \rightarrow \mathcal{T}_1 \rightarrow \cdots \rightarrow \mathcal{T}_L$ form a directed system of tree maps (semigroupoid arrows) that never split a cylinder. This is the precise content of “the MLP gradually mods out”.

3.3 How fast a random He network mods out

At infinite width, two inputs with correlation c have layer- ℓ correlation $\varrho^{\circ\ell}(c)$, with $\varrho(c) = \frac{c}{2} + \frac{1}{\pi}(\sqrt{1-c^2} + c \arcsin c)$ (NNGP recursion; notes I–II). With chord distance $s = \sqrt{2(1-c)}$:

$$1 - \varrho(1-x) = x - \frac{2\sqrt{2}}{3\pi}x^{3/2} + O(x^{5/2}), \quad 1 - \varrho^{\circ\ell}(0) \sim \frac{9\pi^2}{2\ell^2}.$$

Proposition 3.3 (critical stretch: keep fine scales, flatten coarse ones). *Fine separations ($x \rightarrow 0$) are preserved to first order, with relative contraction $O(x^{1/2})$ per layer. Every coarse separation is driven toward the common scale $s_\ell \approx \sqrt{9\pi^2/\ell^2} = 3\pi/\ell$. So the network’s Michon tree is the input tree with its top compressed: coarse cylinders become nearly equidistant at scale s_ℓ , and then merge under ε -chaining for $\varepsilon \gtrsim s_\ell$. Fine structure survives.*

Numerics ($n = 512$, $L = 16$). The input set is $x(\xi) = \sum_k r^{k-1}g_k(\xi_k)$ with $r = 0.55$, $K = 8$ binary digits and random Gaussian $g_k(0), g_k(1)$: 256 inputs forming a Cantor-like set. The table gives the chord distance for pairs whose first differing digit is k , observed and predicted from $\varrho^{\circ\ell}$:

first differing digit k	input	layer 4 obs	layer 4 pred	layer 16 obs	layer 16 pred
1 (coarsest)	1.295	0.809	0.767	0.350	0.369
2	0.697	0.533	0.523	0.285	0.305
3	0.398	0.356	0.337	0.215	0.232
4	0.211	0.186	0.193	0.133	0.154
5	0.116	0.110	0.111	0.088	0.097
6	0.066	0.066	0.064	0.058	0.059
7	0.035	0.038	0.035	0.032	0.033
8 (finest)	0.018	0.019	0.018	0.016	0.017

The coarse/fine ratio drops from 72 to 22. At fixed $\varepsilon = 0.239$, the number of PB ε -chain classes among the 256 inputs is 8, 8, 8, 8, 7, 5, 2 at layers 0, 1, 2, 4, 8, 12, 16.

4 When does stretch-and-fold expand? A Galton–Watson trichotomy

For an activation ϕ at its variance fixed point, the normalized correlation map $\rho(c)$ has nonnegative Taylor coefficients (squared Hermite coefficients, plus a bias constant) and $\rho(1) = 1$. It is therefore a probability generating function, with mean offspring $\rho'(1) = \chi_1 = \sigma_w^2 \mathbb{E}\phi'(u)^2$, the usual order/chaos parameter.

Proposition 4.1 (trichotomy). • Subcritical ($\chi_1 < 1$, *ordered phase*): *extinction is certain and $\varrho^{\circ\ell}(c) \rightarrow 1$ exponentially. The network mods out everything, fast.*

- Critical ($\chi_1 = 1$; *ReLU at He initialization for every σ_w , since ReLU is homogeneous*): $1 - \varrho^{\circ\ell} \sim C/\ell^2$. *This is polynomial flattening with no gap.*
- Supercritical ($\chi_1 > 1$, *chaotic phase*): $\varrho^{\circ\ell}(c) \rightarrow c^* = q$, *the extinction probability, exponentially at rate $\rho'(q) < 1$. A fraction $1 - q$ of input randomness is never modded out; distinct inputs stay pseudo-independent (PRG-like), and the convergence is gapped.*

The order/chaos picture is the known mean-field theory of signal propagation (Poole et al.; Schoenholz et al.). Its reading as a Galton–Watson process with offspring law given by the dual activation is the observation used here.

system	transformations on the profinite structure	spectral behaviour
Gabber–Galil on $\mathbb{Z}_p^2$	bijective isometries, non-amenable group	uniform gap at all levels
tent map / shift	expansion ($\lambda = 2$)	exponential mixing
$\mathbb{Z}^d$ tilings (BBG/SB)	bijective isometries, amenable group	no gap (note I)
random ReLU, He init	critical stretch-and-fold, merging	polynomial, no gap
chaotic smooth nets	supercritical stretch-and-fold	gapped, residual randomness $1 - q$

5 What this clarifies

1. *The MLP’s role in the object.* It supplies a profinite structure: which inputs it identifies at each scale (Proposition 3.2). It also supplies merging tree maps (its layers) and a state (gain-weighted law). Weights and hidden vectors are coordinates for these, not the object itself.
2. *Why the network itself is not an expander.* Its maps merge and, at He initialization, are exactly critical. This is the abstract reason for note II’s ceiling: the network forgets polynomially, with the Galton–Watson tail.
3. *Where expanders do enter.* They act on the transverse/input side, as isometric families on the profinite structure: Gabber–Galil tail operators (Theorem 2.3) or note I’s small-bias arrows realize the hierarchical diffusion with few explicit maps. Expander walks over input codes give randomness-efficient seeds (expander Chernoff bounds), but they cannot beat note II’s variance ceiling.

Problem 5.1 (a neural Gabber–Galil lemma). Find isometric, measure-preserving transformation families on the *network-induced* trees $\mathcal{T}_\ell$, built from weight data (e.g. sign-flip or Jacobian symmetries of a layer), whose local expansion on each level of $\mathcal{T}_\ell$ can be certified. That would give an expander adapted to what the network actually distinguishes, the natural home for a pseudorandom walk that samples the network’s structure rather than its seed.

A Scripts

`code/gg_levels.py` (Gabber–Galil eigenvalues and tail operator) and `code/modding_out.py` (Cantor-structured inputs through a random He network).